

Stagnone di Marsala digital twin: project resumption and status

Cicero Martins Jr.

To: Prof. Giuseppe Ciraolo, Prof. Mauro De Marchis

8 August 2026

1. Summary

Work on the coupled three-dimensional model resumed on 5 August after a pause since mid-June, due to the rewriting of the manuscripts related to GEE. As we already had several runs done and the skeleton of the paper in place, the main outcome of the first three days was to compile everything and finalise the first draft, in LaTeX and formatted for *Estuarine, Coastal and Shelf Science*.

Then the experimental design changed under us, for the better. Two of the eight cells of the factorial had been failing to run, and tracing that failure to its cause let both complete. Their results reverse the paper's central finding. Sections 2 and 3 below are the new picture; the draft has been rewritten accordingly and is ready for your reading.

2. What changed, and why it matters

The design crosses three factors: wave coupling, bed mobility, and whether seagrass resistance is uniform or distributed from a satellite classification. Until yesterday the two cells combining a mobile bed with no wave coupling could not be obtained. The cause turned out to be numerical rather than physical, the contribution of suspended sediment to the density field, and disabling it let both members run to completion with no other change.

With those two cells in hand the attribution comes out differently. On a fixed bed wave coupling costs 0.035 to 0.042 in Liu–Weisberg skill. On a mobile bed the same treatment *gains* 0.155 and 0.232. Bed mobility behaves symmetrically: without waves it costs 0.071 to 0.075 and makes every one of the 35 drifters worse, in both roughness treatments; with waves it gains 0.119 to 0.199.

Change in Liu–Weisberg skill	On a fixed bed	On a mobile bed
From adding wave coupling	−0.035 to −0.042	+0.155 to +0.232
	Without waves	With waves
From adding bed mobility	−0.071 to −0.075	+0.119 to +0.199

So neither treatment improves prediction on its own, and each is what makes the other work. Applied singly they are the two treatments that make the model worse. The two worst and the two best members of the ensemble are built from the same pair and differ only in whether both are switched on. All eight bootstrap intervals behind those numbers are clear of zero.

Distributed seagrass roughness has moved to third order on this evidence. It changes skill by 0.009 or less in three of its four conditions, and by 0.081 in the fourth, which is on a mobile bed with wave coupling. Its effect on the wave field remains real and correctly ordered by canopy density, reducing near-bed orbital velocity by 14.3% over *Posidonia* against 3.3% over exposed rock, so the parametrisation works. It simply does not move surface transport unless the other two are present.

3. What is unchanged

Two results survive the reanalysis and support the framing. Water level still cannot separate the configurations, and the case is stronger with eight members than with six: anomaly RMSE differs by at most 4.2 mm at any station, against a Lagrangian skill range of 0.469 to 0.705 and endpoint separation of 336 to 625 m. And surface current speed exceeds bed speed by a factor of 1.85 to 2.20 at sub-metre



mean depth, which is the quantitative case for solving in three dimensions, consistent with De Marchis et al. (2012) here.

One earlier number was corrected in the process. The vertical shear ratio had appeared to reach 4.5 in the mobile-bed members; that was an artefact of the same sediment-density coupling suppressing near-bed flow, and the true range across all eight members is the narrower one quoted above.

4. Two questions for you

First, the title. The one in the draft is my current preference, but the other two are live and I would value your view.

1. *Interacting effects of wave coupling and bed mobility on Lagrangian transport in a shallow vegetated lagoon.* In the draft. Conservative and conventional in form.
2. *Wave coupling helps or hurts lagoon transport prediction depending on whether the bed can move.* States the finding rather than the topic, and the conditional form claims no more than the eight cells show.
3. *Transport validation of a shallow lagoon model: interacting effects of wave coupling and bed mobility.* Method first, mechanism second.

Second, and more consequential, whether to submit before obtaining a new bathymetric survey. The configurations that best reproduce observed transport are the ones with a mobile bed, and its critical shear stress for erosion has not been constrained against measured bathymetric change. We are using the old bathymetry, so it is possible that a mobile bed helps because it moves the old morphology towards the present one, in which case the improvement is a compensation rather than a physical result. The manuscript states this openly, and notes that the lagoon bed changes by up to 1.4 m over nine days in isolated cells, which is not a plausible rate. If there is a realistic chance of obtaining new bathymetry soon, I would rather wait and rerun with it than defend the present version in review.

5. Status and plan

The draft is complete across all six sections, with eight figures and three tables, 5950 words against a limit of 8000, and 24 references. Highlights and a cover letter are drafted. Outstanding is a graphical abstract, which *ECSS* requires.

After the holidays, and before leaving for Malta, my intention is to close this paper and begin drafting the remaining themes we discussed: the effect of the multi-decadal *Posidonia* decline on wave attenuation, near-bed exposure and residence time; the operational forecast chain with the WetWise portal, together with the Isola Grande channel reopening scenario; and wave-driven sediment mobilisation with the Trapani canal contribution to lagoon siltation.

The full draft, figures and analysis code are in the project repository, and I can send the compiled PDF on request.

Dipartimento di Ingegneria, Università degli Studi di Palermo · cicero.martinsjr@unipa.it